CHAPTER 2

Rooftop Gardening

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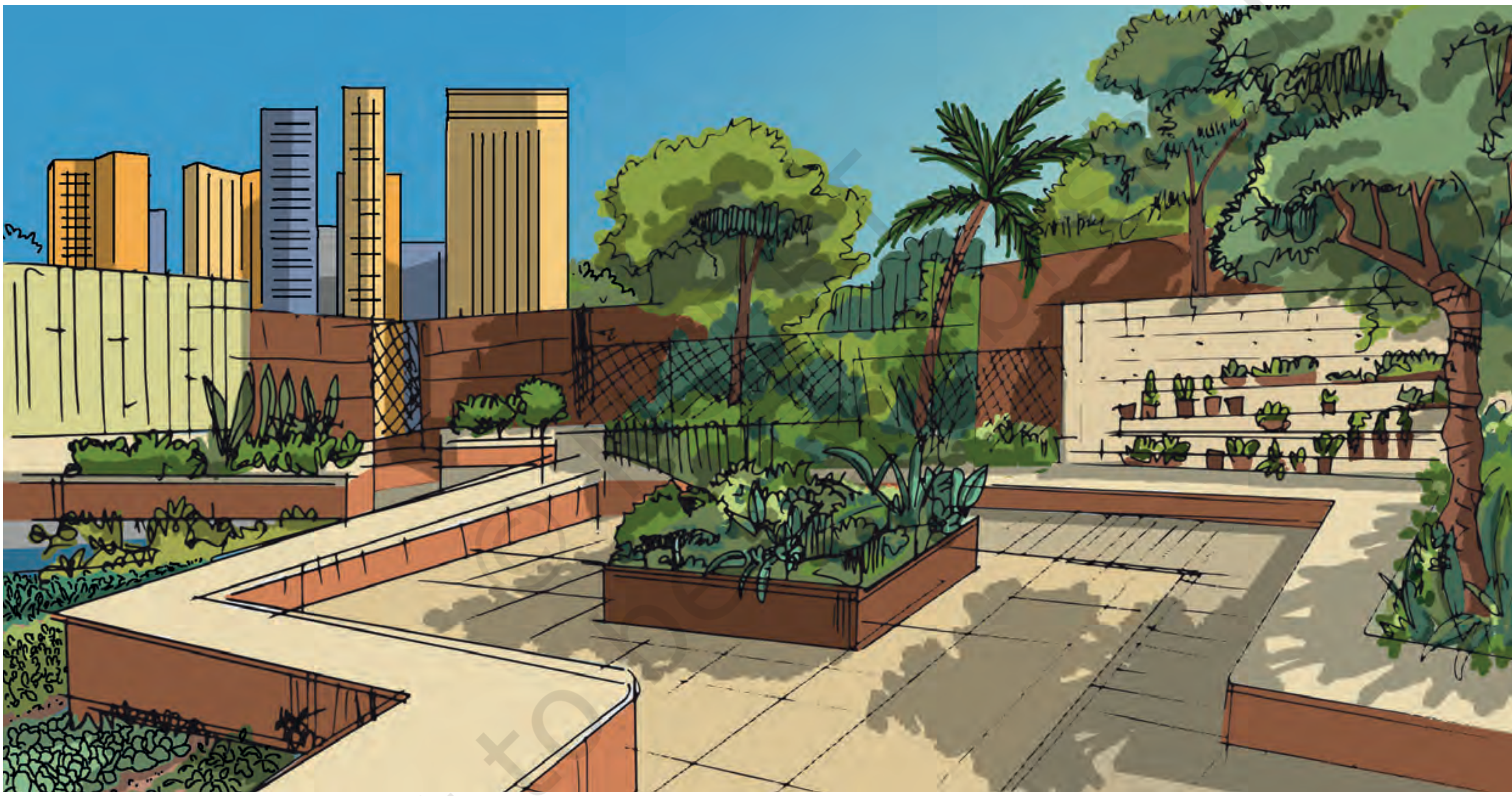

Figure 2.1: *A rooftop garden with shade provided by trees, heavy containers placed at the edge of the roof, a pathway for walking and a rack for pots*

Apart from production of healthy vegetables, rooftop gardening also cools down the temperature, reduces air pollution and creates green spaces that make people feel happy and relaxed. Most importantly, rooftop gardens help people connect with nature and provide a habitat for diverse animals (Figure 2.1).

In this chapter, you will

- Design the layout of a rooftop garden
- Select materials and tools
- Prepare potting mixture
- Conduct a germination test
- Sow, maintain and protect plants
- Harvest, store and package produce

2.1 Introduction

Rooftop gardening involves growing plants on the roof of a house, school or building. Instead of leaving rooftops unused, they can be turned into green spaces, where vegetables, fruits, herbs and flowers are grown in pots, grow bags or specially-prepared beds. This practice is especially useful in cities and towns, where open land for farming is limited. Rooftop gardens reduce dependence on market produce, improve air quality and keep buildings cooler during summers, while making the surroundings beautiful.

This chapter discusses the processes of growing plants in pots, on rooftops, balconies and terraces. The yield of such a garden varies based on the type of plant (leafy greens, fruits or flowers), gardening method used (soil, pots or water-based systems) and care. However, even a small space of just 1 square metre on a rooftop can yield about 10 kg of vegetables in a year.

2.2 Process chart

Defining scope of work

2.2.1 Scoping work

Deciding the scope of the work means that decisions need to be taken regarding the following:

1. Which plants should be grown? Refer to the discussion on agro-climatic relationship in Chapter 1 to identify what you can grow. Ensure that the life cycle of the selected plants is 2–3 months.
2. Other considerations include the amount of sunlight required, the amount of water required (you should select plants that require little water), investment in pots/containers, soil or any other growing media to be

used, weight of pots, and any other specific point brought up by the teacher or expert.

3. What is useful? You need to decide, for example, vegetables for the midday meal or flowers during the festival or wedding season.
4. Where should they be grown? You will also need to decide where to create the garden and how many pots can be placed in the available space.

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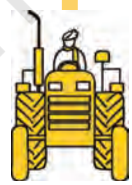
Which plants will you grow? Justify your choice.

2.2.2 Making the process chart

A process chart lists all the tasks you will do along with estimated dates of completion and responsibility (Table 2.1).



Making a process chart

Table 2.1: Template for process chart

Tasks for rooftop gardening	Dates	Responsibility
Layout of rooftop garden		
Soil testing and potting mixture preparation		
Germination test		
Sowing		
Maintenance (irrigation and weeding)		
Protection from pests		
Monitoring growth and providing nutrients		
Harvesting		
Packaging and transport		

2.3 Site visit

To learn from experienced practitioners, visit rooftop gardens with your teacher. Possible site visits can include residences, offices, hospitals, hotels or educational institutes that have developed rooftop gardens.

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Note your observations with reference to Table 2.2.

Table 2.2: Guidelines for observation during site visit

Points of observation/ discussions	Description
Process followed	Key steps and their importance
Tools and materials used	Materials used and their storage Tools used and their maintenance
Safety protocols	Using tools correctly, safety precautions, etc.
Schedules	Frequency and timing of key tasks
Quality criteria	Criteria of quality for inputs, process and output
Technology use	Digital tools/apps used

Think of any other points for observation, while visiting the site. For example:

1. What does the gardener value the most about their work (for example, quality of produce, connect with nature, opportunity for employment, earnings, etc.)?
2. Possible challenges and how to overcome them

Create a process chart for the work you will do.

Essential conditions for growth

2.4 Designing the layout

When designing the layout for rooftop gardening, you must keep the following in mind:

1. **Space:** Adequate space must be left between pots or containers for watering and general maintenance.
2. **Sunlight requirement:** Select the part of the rooftop that receives at least 5–6 hours of direct sunlight every day. Different plants have different needs, so areas with partial shade can be used for herbs and leafy vegetables, while sun-loving plants, like tomatoes and cucurbits can be placed in open areas.
3. **Weight and safety considerations:** The weight of pots/ containers, growing media, and other materials can overload a roof and damage the building. Therefore, heavier containers, such as large pots or cement planters

should be placed close to the edges of the roof or above the beams, where the roof can support more load. This requires an understanding of the structure of the roof (Figure 2.2) and the standardised maximum weight load limit for rooftops.

4. **Water management:** Pots should be placed on trays, tiles or stands, so that excess water is collected and the roof surface is protected from seepage. Proper drainage arrangements prevent waterlogging and extend the life of the roof.

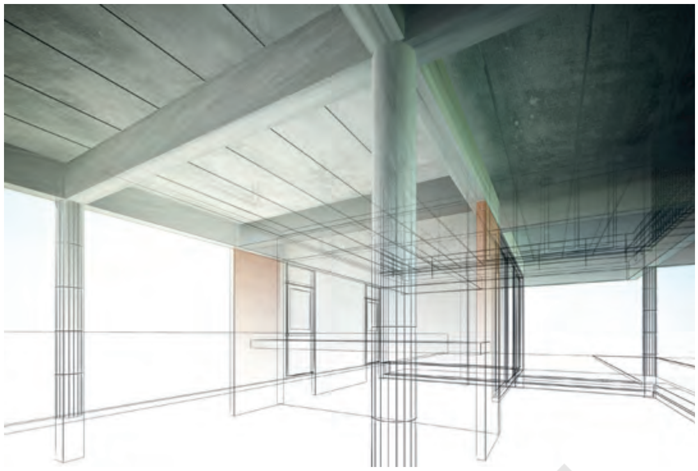

Figure 2.2: *Beams support the roof; more weight can be placed over beams compared to the rest of the roof.*

5. **Provision for shade:** Certain plants cannot tolerate strong, direct sunlight. For these, shaded areas can be created using shade nets, bamboo screens or vertical trellises. This helps in balancing light exposure across the rooftop.
6. **Protection from wind:** In rooftop gardening, plants will need protection from strong winds, which can damage plants or even throw entire pots off-balance. Use instruments from the meteorological observatory that you developed as part of Chapter 1 to record the direction of wind for about a week before finalising the layout.

Essential conditions for growth

TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

You can use 3D modelling software like Computer Aided Design (CAD) to design the layout of the roof. You can also determine the intensity of sunlight on different parts of the roof using Lux Meter mobile app.

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Draw and label a detailed sketch of the layout of your rooftop garden. Give reasons for your decisions related to the placement of pots.

CHECK YOUR UNDERSTANDING

Figure 2.3: *Representation of a rooftop garden*

On the basis of Figures 2.3 and 2.4, answer the following questions:

1. Why do you think some pots are placed in area B? Can you give examples of such plants from your geographic region?
2. Do you think the pots placed in area C can be placed in area E? Why or why not?
3. If the number of plants is increased, which area of the rooftop garden should be expanded first and why?
4. Do you think the layout has enough space to accommodate all the plants? Can you imagine any other layout?

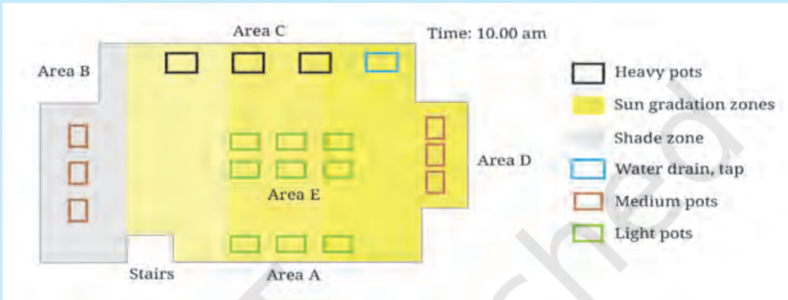

Figure 2.4: *Sketch of layout of rooftop garden*

CASELET

Structural Safety and Waterproofing

Students of Grade 9 in Government High School formed the 'Go-Green' team to develop a rooftop garden (Figure 2.5). To ensure safety, they first ensured the parapet wall was strong. Next, they consulted the contractor, who had built the school building. He informed them that the roof could safely support a maximum load of 150 kg per square meter.

Figure 2.5: *Setting up a rooftop garden*

The students selected an area of 4 m × 3 m (12 m²) for their garden, which meant the total permissible load on that section of the roof was 1,800 kg. After consultation with their teacher and experts, they estimated that the combined weight of around 50 pots, the potting mixture and other components of the rooftop garden would be about 1,200 kg – well within the safe limit.

With this confirmation, they waterproofed the roof with tarpaulin liner and installed support of bamboo to support climbers.

2.5 Selecting materials

After the layout has been planned, the next step is to select materials. The table below (Table 2.3) will help you make decisions for gathering materials for the rooftop garden. You may recall some of these from Grades 6 to 8 *Kaushal Bodh*.

Table 2.3: Selection of materials

Materials	Options	Safety note
Pots/ containers	Terracotta pots (allow air and moisture to pass, preventing waterlogging around roots), plastic pots (lightweight), grow bags (flexible and easy to move), cement pots (limited use since heavy and retain heat), wooden containers (allow air to pass, preventing temperature fluctuations and waterlogging) and recycled material (for example, old buckets with holes drilled in bottom, old sinks)	Place heavy pots/containers only near beams or roof edges. Distribute weight evenly across the roof
Potting mixture	Soil, compost mixtures and cocopeat/ perlite/vermiculite	Avoid using only soil, as it retains excess water and increases the load on the roof
For watering	Watering cans, small sprinklers and hose pipe attached to taps, and trays/ saucers to prevent seepage	Always place trays/saucers under pots to collect extra water and prevent seepage
Shade and support structures	Shade net, bamboo screens and lightweight movable trellises	For providing shade and for climbers
Gloves	-	Always wear gloves while handling soil and organic matter
Other components	Compost bin and storage area	Care to be taken to avoid infestation by rats and/or pests, and proper storage with lock arrangements

SAFETY

Safety protocols/instructions to be followed while lifting and carrying load, and handling tools and materials. Protective gear to be used as required.

Select tools and materials

2.6 Selecting tools for doing work

Table 2.4 provides a list of tools necessary for developing the rooftop garden, along with their use and safe handling. You may recall some of these from the Grades 6–8 *Kaushal Bodh*.

Table 2.4: Selection of tools

Tools	Use	Safety note
Hand trowel	Digging soil, planting seeds and transplanting seedlings	Do not leave sharp tools lying around; store them safely after use
Pruning scissors	Cutting stems, removing dried leaves and shaping plants	Handle carefully; keep blades closed when not in use
Watering can/ spray bottle	Watering plants gently, especially seedlings	Avoid spilling water on the floor; keep a firm grip to prevent slipping
Buckets/trays	Carrying soil mix and collecting water drainage from pots	Do not overload buckets; lift small quantities to avoid strain
Pots/grow bags/ grow beds and nursery trays	Growing seedlings and plants	Choose plant container as per need and convenience in handling; avoid large pots as they will be difficult to handle

Cost estimation and documentation

2.7 Making Bill of Materials

The Bill of Materials (BoM) helps in estimating costs in advance, avoiding wastage by buying only what is necessary and organising the work step by step. You can use the template in Table 2.5 or modify it to list your estimated materials and tools. Cost of labour should also include the estimated time spent in doing the work.

CASELET

The 'Go-Green' team was almost ready to begin developing their rooftop garden, but felt a bit unsure about the quantity and cost of various materials required. Their teacher advised them to prepare a Bill of Materials for the project. To understand the concept better, the students sought guidance from a parent, who is an entrepreneur. Table 2.5 shows the Bill of Materials they prepared.

Table 2.5: Bill of Materials

Items	Quantity	Estimated cost (in ₹)	Remarks (if any)
Terracotta pots	15	500	As per the space available on roof/ convenience of use.
Hose pipe	10 m	600	Length calculated as per distance between tap and most distant pot
Bamboo for safety railing and trellis	3 pieces	250	Length of each pole is 5 m
Potting mix	100 kg	-	Donated by the local nursery
Cost of labour	**Value (Time in hours × hourly estimate × frequency per week)**	**Estimated cost (in ₹)**	**Remarks (if any)**
Watering plants	0.5 × ₹ 25 × 4	50	
Care and mainte-nance	0.5 × ₹ 25 × 2	25	
Total		**1425**	

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Prepare a Bill of Materials for your rooftop garden.

2.8 Pot preparation and sowing

You may have learned the basics of soil preparation and sowing in Grade 6 *Kaushal Bodh*. This section contains guidelines specifically for rooftop gardening.

You will need to prepare a potting mixture, for the rooftop garden as shown in Figure 2.6. For preparing a potting mixture, follow the steps below:

1. Collect material for the potting mix, keeping in mind the weight (lightweight is better) and nutrient supply (manure/compost).
2. Mix cocopeat/ vermiculite/ perlite with compost/ vermicompost and soil in 1:1:1 proportion.
3. Test for pH as described in Chapter 1.

Preparing growing media

DID YOU KNOW?

Low weight but nutrient rich potting mixture

Potting mix (please refer to Grade 6 *Kaushal Bodh*) is used, because ordinary garden soil is too dense and heavy for pots. When watered, soil particles get packed together tightly, which prevents air from reaching the plant roots. This causes poor drainage, leading to waterlogging and rotting of the root.

Potting mix is specifically designed to remain loose and airy. This ensures better aeration (oxygen for roots) and allows excess water to drain out easily, creating an ideal environment for healthy plant growth in a pot or container, while reducing the weight on the roof.

For example, a potting mixture of cocopeat + compost + perlite in 1:1:1 proportion will weigh around 290 g for a pot of 1 L capacity. If you replace perlite with soil, the same mixture will weigh around 700 g because soil is much heavier than perlite. So, just by replacing or reducing the amount of soil, you can reduce the weight of pots.

Figure 2.6: *Potting mixture weight distribution*

TASK

Testing pH of potting mixture and adjusting pH for 20 pots/ containers of 1 L volume

1. Follow the pH testing process given in Chapter 1.
2. If pH is acidic (below 6.0), add garden lime (calcium carbonate) and if pH is alkaline (above 8.5), add sulphur or organic manure. Quantity of lime or sulphur can be determined through trial and error, for example, adding one tablespoon of garden lime or sulphur/organic manure and testing pH again till it is satisfactory.
3. Test pH again and adjust it to 6.5 to 7.0 for the potting mix.

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Note the details of your work so far; you can use the template in Table 2.6.

Table 2.6: Preparation of growing media

Plants you are growing	Method of sowing	Materials for potting mix, and proportion used	Result of pH testing of potting mixture and amendments

DID YOU KNOW?

Hybrid seeds are specially produced for higher yield. These seeds are costly and can be used only once. Farmers preserve indigenous knowledge related to harvesting and storage of local seeds. This is very important for 'seed security' and biodiversity conservation. Seeds are harvested carefully and preserved for the next season.

Seed selection

TASK

Testing seed germination rate

You can check the quality of seeds by testing their germination rate. Seed germination rate is also printed on seed packets, but it may vary in the field as per sowing conditions (for example, soil moisture and temperature).

You may have done a seed germination test in Grade 7 *Kaushal Bodh*. Now, you can test the seeds in a potting mix to verify germination rate following the steps in Figure 2.7.

1. Count the seeds (you may recall that Grade 6 *Kaushal Bodh* recommends reading seed packets or asking experts about planting depth and spacing instructions; you can estimate the number of seeds required based on the size of the pots/ containers).

2. Plant them in a seedling tray, pot or container.

3. Water regularly.

4. After 2–3 weeks, count the number of germinated seeds and compare them to the number of seeds planted.

5. Calculate germination percentage and compare it with the 'seed packet' claim, if possible.

6. You can use these seedlings for your rooftop garden by carefully transplanting them to pots or containers.

Figure 2.7: *Process to test seed germination rate*

You have learned the basics of seed sowing in Grades 6–8 *Kaushal Bodh*. In case of a rooftop garden, do remember to protect young seedlings from strong winds, using small covers or nets, or by placing pots against a wall.

Maintenance and monitoring

2.9 Maintenance and recording growth of plants

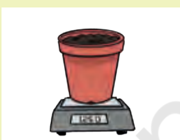
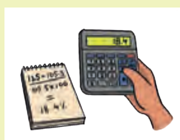
2.9.1 Irrigation

If you provide less water, plants will not grow properly, due to water stress (when demand for water exceeds available supply), while heavy watering may lead to suffocation of roots. In soil-based farming, roots can grow deep in the soil in search of water. Furthermore, excess water can percolate down without harming the roots. As both these advantages are not available in pots/containers, you must irrigate with care.

TASK

Estimating water requirement

To estimate the approximate amount of water required by your garden, you will need to calculate the Water Holding Capacity (WHC) of the potting mixture. WHC is the maximum amount of water that soil can hold; in this case, soil has been replaced by potting mixture (Figure 2.8).

	1. Add 500 g potting mixture to the container/pot.		2. Note the weight of the potting mixture, along with the container/pot (dry weight).
	3. Slowly add water until the potting mixture gets saturated (does not absorb any more water).		4. Let the water drain out (this will take 10–15 minutes).
	5. Once the water stops draining out, note the weight of the pot again (wet weight).		6. Calculate WHC = (Wet weight – Dry weight)/(Dry weight) × 100

For example, $\text{WHC} = \dfrac{130-105}{105} \times 100 \approx 24\%$

Figure 2.8: *Process to calculate water holding capacity*

On the basis of the WHC of the potting mixture, you can calculate the approximate water requirement of your garden. For example, if you have used 400 kg of potting mix and your WHC is 25 per cent then you need 100 L water to irrigate your garden. But do remember that this is an approximate calculation. Also, this does not mean that you need to use 100 L of water per day. This is the maximum water required by plants for growth.

Maintenance and monitoring

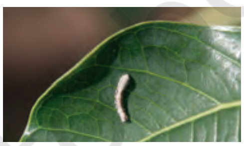
2.9.2 Plant protection

2.9.2.1 Protection from diseases and pests

Plants need protection from diseases and pests. You cannot eliminate all diseases and pests, as they are part of our ecology. However, you can limit damage to crops. The integrated disease or pest management approach involves physical (for example, removal of diseased leaves and insect larvae), chemical (for example, spraying organic pesticide) and biological (for example, rearing beneficial insects) methods.

To protect your crop, you will need to observe each plant carefully for symptoms of disease or pest infection (for example, presence of spots or insect bites). You can identify the disease or pest, using a mobile app or by sending photographs to experts. Refer to Table 2.7 to identify the actions to be taken.

Table 2.7: Examples of common pest infestation and remedies

Symptoms/damage	Pest	Plants likely to be infested	Remedy
Chewed leaf 	Caterpillar 	Cabbage, cauliflower, broccoli, tomato, brinjal and chilli	*Neem* spray and turmeric to repel
Discoloured leaves 	Spider mites 	Rose, hibiscus, beans, brinjal, cucumber, watermelon and indoor plants	Spray with water to dislodge pests
Deformed leaves 	Aphids 	Mustard, cabbage, cauliflower, tomato, brinjal, chilli, rose, hibiscus and marigold	*Neem* spray, chilli garlic spray and manual removal at early stage

Large holes and shiny trails 	Slug/Snail 	Cabbage, lettuce, spinach and seedlings of all vegetables	Manual removal and salt lines
Wilted and chewed leaves 	Chafers (beetles) 	Rose, hibiscus, jasmine, potato, groundnut, maize and sugarcane	Digging to expose larvae, neem spray and hand picking beetles
Black mould and weak sticky leaves 	White fly 	Tomato, brinjal, chilli, cotton, okra (bhindi), cucumber and hibiscus	*Neem* spray and soap water spray

Maintenance and monitoring

2.9.2.2 Protection from weeds

In simple terms, weeds are unwanted plants in your garden. Farmers remove weeds by physical means (for example, hand-weeding, using power or mechanised weeders), through mulching (for example, organic mulching using materials, like bark, wood chips, straw, hay, grass clippings and shredded leaves or plastic paper) (Figure 2.9) and also through spray chemical weedicides. Refer to Figure 2.10 to manually remove weeds from your pots/containers.

(a) (b)

Figure 2.9: *(a) Mulching using plastic sheets and (b) organic mulching*

1. Identify the weeds, growing near the main plant.

2. Loosen the soil around the weed gently, using your fingers, so that the roots can come out easily.

3. Hold the weed firmly at the base (near the soil surface).

4. Pull the weed out slowly and steadily, making sure the complete root comes out.

5. Level the soil around the main plant, after removing the weed.

Figure 2.10: *Manual weeding*

2.10 Harvesting and storage

The timely collection of produce is very important, because it ensures both quality and a longer shelf life. If harvesting is delayed, the produce may become overripe, lose taste or spoil quickly (Figure 2.11).

You have already learnt the basics of harvesting in Grade 6 and 8 *Kaushal Bodh*. Some important points to keep in mind for rooftop gardening are:

1. **Gentle handling:** Since rooftop gardens usually produce small quantities, every fruit, leaf or flower is valuable. Handle them carefully to avoid bruising or crushing.
2. **Right time:** Harvesting in the early morning or late evening keeps the produce fresh and reduces wilting. In the heat of the day, plants lose more water and become limp.
3. **Use of tools:** Simple tools, like scissors, pruning shears or small knives, can make harvesting easier and cleaner. Pulling the produce by hand may damage the plant.
4. **Stage of maturity:** Different crops have different maturity indicators. For example, spinach leaves should be picked when tender, tomatoes when they turn red and coriander before flowering.

Harvesting, packaging and storing

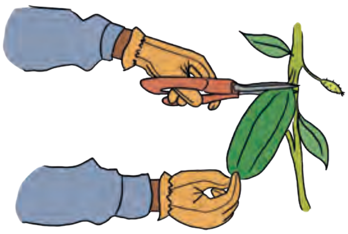


Figure 2.11: *Harvesting vegetables*

Storage of produce

Storage is just as important as harvesting, because it determines how long the produce will stay usable.

1. **Cleaning:** Wash vegetables and herbs gently to remove soil and dust. Dry them before storing.
2. **Containers:** Use baskets or perforated containers, that allow air to circulate.
3. **Temperature:** Leafy vegetables, like spinach and coriander stay fresh longer if kept in a cool, moist place or wrapped in damp cloth. Fruits like tomatoes should be stored at room temperature until fully ripe.

TASK

Observe how long the produce stays fresh in different storage conditions (for example, room temperature, refrigerator or wrapped in cloth).

Table 2.8 lists the time from sowing to harvest and appropriate storage methods for common rooftop crops.

Table 2.8: Harvest time and storage method of crops

Crop	Harvest time (in days)	Storage method
Lettuce	45–55	Store in a plastic bag in the refrigerator for up to 1 week
Spinach	30–60	Wash, dry thoroughly and store in an airtight container in the refrigerator for less than 1 week
Radish	25–45	Remove the leaves and refrigerate
Okra (Bhindi)	45–55	Store in a container and refrigerate
Capsicum	60–80	Refrigerate for 1–2 weeks
Basil, mint, coriander	25–30	Place stems in a jar of water on the counter or loosely wrap in a damp paper towel
Tomato	55–70	Keep at room temperature
Brinjal	60–70	Store in a cool place.

2.11 Packaging and transport

Packaging is important to keep the produce fresh for a longer duration. Common processes for packaging produce include (Figure 2.12) the following:

1. **Leafy vegetables (like spinach, coriander, mint):** Wrap in a clean, damp cloth or newspaper to prevent wilting. Refrigerate soon after.

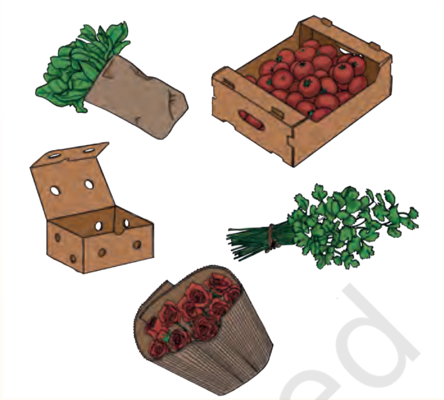


Figure 2.12: *Types of packaging*

2. **Fruits like tomatoes and chillies:** Store in baskets, trays or ventilated corrugated cardboard boxes. Do not pack in tight plastic bags with no aeration, as these trap moisture and cause rotting. Many fruits naturally emit ethylene gas. Corrugated boxes with holes or baskets that allow air to circulate are used for fruit. They help to remove ethylene gas, which causes fruit to ripen faster. By removing this gas, fruits can be stored for a longer period, especially during transportation.
3. **Herbs:** Tie small bunches neatly with jute string or paper bands. Label them with names for easy identification.
4. **Flowers:** Place in baskets lined with damp cloth or corrugated sheets. The moisture keeps the flowers fresh for a longer time.

2.12 While developing the rooftop garden

1. Follow safety protocols related to lifting, and handling of tools and materials. Ensure safety of all, while working on rooftops.
2. Plants must be selected considering local agro-climatic conditions (rainfall, temperature, humidity, snowfall, etc). Locally-available varieties grow well, help in improving air quality and cool building, along with greater chances of a good harvest.
3. Consider design factors, like availability of sunlight, space, structural safety, water management and shade provision, while planning the rooftop garden.
4. Use lightweight pots and potting mixture test the potting mixture and make amendments as needed.
5. Test seed germination rate to try and ensure growth of all planted seeds.
6. Set watering schedules, regularly monitor plants for pest and diseases, and use organic pesticides for healthy plant growth.
7. Harvest at the right time, using simple methods to preserve and package produce for distribution.

2.13 Assess your learning

1. During a site visit, students notice that one rooftop garden uses grow bags, while another uses clay pots.
   i. List one advantage and one disadvantage of each container.
   ii. Which one would you recommend for your school rooftop garden and why?

2. You have prepared two potting mixes as follows:
   i. Mix A: Soil + Compost + Cocopeat
   ii. Mix B: Soil only

   After one month, plants grown in the Mix A potting mix look healthier and are lighter to carry, while plants in Mix B pot are heavy and waterlogged.

   Explain why Mix A is better for rooftop gardening.

3. A group of students noticed their tomato plants turning yellow despite regular watering. Later, they found out that the pots were waterlogged due to blocked drainage holes.
   i. What went wrong?
   ii. Suggest two preventive measures for the future.

4. A group of students harvests 5 kg of spinach, packs half in plastic bags and half in damp cloth. After two days, the spinach in a damp cloth is fresher.
   i. Why did the spinach in a damp cloth stay fresh longer?
   ii. What lesson does this give about packaging?

5. Your school has received a donation of 15 (6 inch) pots for rooftop gardening as well as spinach, tomatoes and marigold seeds.
   i. Which plants will you select?
   ii. Justify your choice based on agro-climatic conditions, usefulness (food/decoration), life cycle and water requirement.

6. A family wants to try rooftop gardening, but is worried about the cost.
   i. Suggest two recycled materials they can use as containers.
   ii. How does recycling support the environment?

7. Of the tasks that you did, which did you enjoy the most? Which did you enjoy the least? Give examples of what went well and what did not go well. What would you do differently next time?

8. Give examples of how you can apply your learning in a real-life situation.